Wire Ampacity Chart

Allowable ampacity for copper and aluminum conductors at the three insulation temperature ratings, and the factors that reduce it.

Ampacity is the current a conductor can carry continuously without exceeding its temperature rating, and it depends on the insulation rating as much as on the wire size. A 6 AWG copper conductor is good for 55 A at 60°C, 65 A at 75°C and 75 A at 90°C. Those figures hold for not more than three current-carrying conductors together at an ambient of 30°C (86°F). Anything hotter or more crowded reduces them.

Copper

Size	60°C	75°C	90°C
14 AWG	15	20	25
12 AWG	20	25	30
10 AWG	30	35	40
8 AWG	40	50	55
6 AWG	55	65	75
4 AWG	70	85	95
3 AWG	85	100	115
2 AWG	95	115	130
1 AWG	110	130	145
1/0 AWG	125	150	170
2/0 AWG	145	175	195
3/0 AWG	165	200	225
4/0 AWG	195	230	260
250 kcmil	215	255	290
300 kcmil	240	285	320
350 kcmil	260	310	350
400 kcmil	280	335	380
500 kcmil	320	380	430

Amperes. Not more than three current-carrying conductors in a raceway, cable or directly buried, at 30°C ambient.

Aluminum and copper-clad aluminum

Size	60°C	75°C	90°C
14 AWG	—	—	—
12 AWG	15	20	25
10 AWG	25	30	35
8 AWG	35	40	45
6 AWG	40	50	55
4 AWG	55	65	75
3 AWG	65	75	85

Size	60°C	75°C	90°C
2 AWG	75	90	100
1 AWG	85	100	115
1/0 AWG	100	120	135
2/0 AWG	115	135	150
3/0 AWG	130	155	175
4/0 AWG	150	180	205
250 kcmil	170	205	230
300 kcmil	195	230	260
350 kcmil	210	250	280
400 kcmil	225	270	305
500 kcmil	260	310	350

Amperes, on the same basis. A dash means the size is not used in that material.

Ambient temperature correction

Ambient	60°C insulation	75°C insulation	90°C insulation
30°C (86°F)	1.00	1.00	1.00
35°C (95°F)	0.91	0.94	0.96
40°C (104°F)	0.82	0.88	0.91
45°C (113°F)	0.71	0.82	0.87
50°C (122°F)	0.58	0.75	0.82
55°C (131°F)	0.41	0.67	0.76
60°C (140°F)	—	0.58	0.71

Multiply the ampacity by the factor for the insulation rating when the ambient is above 30°C. A dash means the conductor may not be used at that ambient.

More than three current-carrying conductors

Current-carrying conductors	Percent of ampacity
4 to 6	80%
7 to 9	70%
10 to 20	50%
21 to 30	45%
31 to 40	40%
41 and above	35%

Multiply the ampacity by this percentage when more than three current-carrying conductors share a raceway or cable.

Where the values come from

The ampacities correspond to NEC Table 310.16, the ambient correction factors to Table 310.15(B)(1), and the adjustment factors to Table 310.15(C)(1). They are reproduced here for reference and are not computed by Voltfield. Editions of the code differ, table numbers have changed between editions, and local amendments apply, so check any value you intend to rely on against the edition your jurisdiction has adopted.

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What this chart does not tell you

The chart covers conductors in a raceway, cable or directly buried, up to 500 kcmil. It does not cover conductors in free air, which carry more; cables in tray, flexible cords, or conductors rated above 2,000 V, which have their own tables; or the rooftop and other special conditions the code treats separately. It does not tell you the overcurrent device, and it does not check voltage drop, which on a long run sets the size before ampacity does.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/wire-ampacity-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.